

SBC 201 Chapter 18 Soils and Foundations — Group R-2 High-Rise Dimensional Requirements Matrix (2024)

1. Document metadata and use limitation

- **Project basis:** Riyadh, Saudi Arabia; Group R-2 residential high-rise; an occupied floor is stated to be more than 23 m above the relevant reference level.
- **Deliverable tier:** Project-use matrices in Sections 1–6 (design-check rows, not pasted inventory), plus a coverage summary and unresolved-source register. The full row inventory is not published.
- **Code/source basis:** SBC 201 (2024), Chapter 18, source file Reference\SBC 201 2024\source_reference\Chapter_18 – SOILS AND FOUNDATIONS.txt.
- **Extraction audit:** Skill-run. Chapter 18 is a pointer chapter: the charging sentence sends soils and foundations to **SBC 303**; the only table is a flattened OCR cross-reference map. Internal inventory: **0** independently checkable numeric records. The flattened map is a non-numeric **Verify source** hold and is not a design-release pairing.
- **Model:** Cursor Grok 4.6.
- **Prepared:** 2026-08-27.
- **Status:** Source-only architectural advisory matrix for design coordination. It is not a stamped compliance statement, geotechnical report, structural design, SCD NOC, SBPS approval, legal opinion, or permission to construct.
- **Outbound-source rule:** No value in this matrix has been imported from SBC 303, Chapter 16, Chapter 17, commentary, IBC memory, or the existing chapter summary. Where Chapter 18 sends the user to SBC 303, this matrix records the dependency without supplying the outbound value.

Scope and assumptions

1. Group R-2 and high-rise status are project statements, not independently verified classifications.
2. The exact Riyadh AHJ/permit pathway, project stage and SCD NOC status are unconfirmed; therefore this matrix does not conclude compliance.
3. This attached chapter **does not contain** soils, excavation, groundwater, retaining-wall, shallow-foundation or deep-foundation criteria. Those requirements are charged to **SBC 303**.
4. SBC 303 is not attached to this extract. No bearing, frost, pile, drainage, excavation or retaining number is adopted.
5. Foundation type, basement/podium extent, groundwater and geotechnical investigation status are unconfirmed.
6. The Chapter 18 cross-reference table is concatenated OCR. No reconstructed 18xx-to-SBC 303 pairing is adopted as a design-release map.

2. Legends

Applicability

Status	Meaning
Direct	Expected to govern the stated R-2 tower basis, subject to confirmed geometry and design data.
Conditional	Governs only when the stated feature, load, foundation type or mixed-use/podium condition exists.
Not typical	Unrelated occupancy-only rule; omitted from this deliverable unless the gap register already opened that use.
External verification	Chapter 18 points to another section/code/standard, or the project/AHJ basis must be confirmed before use.

Source confidence

Status	Meaning
Verified	Requirement and any stated numeric value were checked against unambiguous mandatory Chapter 18 source text.
Verify source	OCR, flattened table, page-split, or footnote attachment is unresolved. Not a design-release value.

3. Project decision and gap register

Decision / gap	Current project basis	Why it controls Chapter 18 application	Required project action
SBC 303 extract	Not attached	Chapter 18 contains no soils or foundation criteria; all technical provisions live in SBC 303	Obtain the applicable SBC 303 edition and commission a source-only extract before foundation freeze
Geotechnical investigation	Unconfirmed	SBC 303 design cannot start without site soils, groundwater and bearing data	Engage a licensed geotechnical engineer; issue a signed investigation for the tower footprint and any podium/basement
Foundation type	Unconfirmed: shallow vs deep	Selects which SBC 303 chapters/tables apply once the published Chapter 18 map is verified	Structural engineer to lock foundation type against the geotech report; do not assume a type from this chapter
Basement / podium excavation	Unconfirmed	Excavation, retaining and groundwater control are SBC 303 problems, not Chapter 18 numbers	Freeze basement/podium extents, adjacent-property setbacks and dewatering strategy with structure and geotech
Groundwater / dampproofing	Unconfirmed	Waterproofing and drainage of below-grade walls/floors are not quantified in this chapter	Record design water table and below-grade envelope strategy in the structural/geotech package
AHJ / permit pathway	Unconfirmed	Foundation permits and inspection hold points are not defined by this pointer chapter	Qualified local consultant to confirm Baladiya/SBPS and any SCD interface for below-grade construction

4. Mandatory pointer

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
Chapter 18 charging	Soils and foundations pointer	Requirements of soils and foundations shall be in accordance with SBC 303	All soils, excavation and foundation work	None stated	External verification	Commission geotechnical investigation and structural foundation design to SBC 303; do not size basements, podiums or bearing from this chapter	Verified

5. Chapter 18 to SBC 303 map (OCR hold)

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
Chapter 18 cross-reference table	SBC 201 to SBC 303 map	Flattened OCR table; no reconstructed 18xx-to- SBC 303 pairing is adopted	Any SBC 201 reference to a Chapter 18 section or table	Concatenated cells; trailing period on 1803.5.5.; duplicate 1803 row	External verification	Obtain the published two-column map before routing any Chapter 18 cite to an SBC 303 destination	Verify source

6. Project-use controls

1. Use the **Verified** charging row as the only Chapter 18 design instruction: soils and foundations follow **SBC 303**.
2. Treat the **Verify source** cross-reference row as a hold point. Do not place any reconstructed 18xx-to-SBC 303 pairing on issued-for-approval drawings.
3. Do not import bearing pressures, frost depths, pile capacities, excavation limits, retaining-wall dimensions or drainage values from SBC 303, IBC memory or the chapter summary.
4. Record geotech, foundation-type and basement/podium decisions in the project Golden Thread; this matrix is not evidence of SCD NOC or stamped compliance.

7. Coverage summary

Internal inventory of the attached Chapter 18 extract (numbered code, exceptions, tables, footnotes; commentary excluded). Row-level records are not published.

- **Inventory scope:** numbered code, exceptions, tables, footnotes (commentary excluded)
- **Total independently checkable numeric records:** 0
- **Verified:** 0
- **Verify source:** 0
- **Non-numeric holds:** 1 flattened cross-reference table (see unresolved-source register)

Coverage cross-check against SBC 201 Chapter 18 Soils and Foundations (2024)_CS.md was topics-only: mandatory SBC 303 pointer; existence of a Chapter 18 to SBC 303 section map. No CS.md pairing or value was copied into a matrix cell.

8. Unresolved-source register

Hold points for OCR / flattened-table items that appear in project-use as **Verify source**. No value in this register is a design-release figure.

Affected table / clause	Why unverified	Control note
Chapter 18 cross-reference table	Concatenated two-column OCR; no cell separators; trailing period on 1803.5.5.; 1803 appears twice	Obtain the published two-column map before using any 18xx to SBC 303 pairing; do not adopt reconstructed destinations